



Cambridge Assessment Admissions Testing

Test of Mathematics for University Admission

Paper 1 2017 worked answers

Test of Mathematics for University Admission
2017 Paper 1
Worked Solutions

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

Question 1

We are given the derivative of y , and we want to find y itself, so we need to integrate. We start by expanding the fraction:

$$\begin{aligned}\frac{dy}{dx} &= 3x^2 - \frac{2-3x}{x^3} \\ &= 3x^2 - 2x^{-3} + 3x^{-2}\end{aligned}$$

so we can now integrate term-by-term to get

$$y = x^3 + x^{-2} - 3x^{-1} + c$$

Now as $y = 5$ when $x = 1$, we can substitute to give

$$5 = 1^3 + 1^{-2} - 3 \times 1^{-1} + c = -1 + c$$

so $c = 6$ and the answer is C.

Commentary: It turns out that there is a subtlety in this question which we usually ignore at school level. The question assumes that there is a ‘simple’ formula for y in terms of x . Since the function is not defined at $x = 0$, it is possible that we have different constants when $x > 0$ and when $x < 0$, so we could have, for example,

$$y = \begin{cases} x^3 + x^{-2} - 3x^{-1} + 6 & \text{when } x > 0 \\ x^3 + x^{-2} - 3x^{-1} + 8 & \text{when } x < 0 \end{cases}$$

This may seem quite strange, but it would be a valid solution to the differential equation given in the question.

Question 2

We expand the brackets, writing everything as powers of x :

$$\begin{aligned} f(x) &= \left(\frac{2}{x} - \frac{1}{2x^2} \right)^2 \\ &= \frac{4}{x^2} - 2 \frac{2}{x} \frac{1}{2x^2} + \frac{1}{4x^4} \\ &= 4x^{-2} - 2x^{-3} + \frac{1}{4}x^{-4} \end{aligned}$$

(Take care with the last simplification: writing $\frac{1}{4x^4} = 4x^{-4}$ is a common error!)

Then differentiating gives

$$f'(x) = -8x^{-3} + 6x^{-4} - x^{-5}$$

and differentiating a second time then gives

$$f''(x) = 24x^{-4} - 24x^{-5} + 5x^{-6}$$

We can now substitute $x = 1$ to give

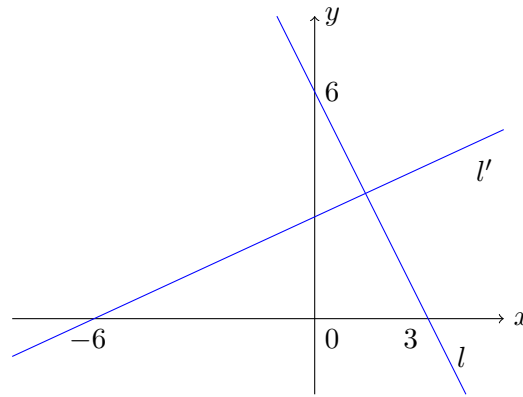
$$f''(1) = 24 - 24 + 5 = 5$$

so the correct answer is C.

Question 3

It is always worth drawing a sketch for questions like this, to ensure that we know what area is being referred to.

The line l has a negative gradient, and passes through the point $(0, 6)$; it also intersects the x -axis at $(3, 0)$. So the sketch looks something like this:



We see therefore that the required area is a triangle with base length along the x -axis of $3 - (-6) = 9$. Thus to find the area we only need to find the y -coordinate of the point of intersection of the two lines.

The equation of the second line, l' , can be given by $y - y_1 = m(x - x_1)$: its gradient is the gradient perpendicular to -2 , which is $\frac{1}{2}$, and it passes through $(-6, 0)$, so it has equation

$$l' : y - 0 = \frac{1}{2}(x - (-6))$$

which simplifies to $y = \frac{1}{2}x + 3$. We can find the coordinates of intersection by solving the equations of l and l' simultaneously. Substituting for y , we get

$$6 - 2x = \frac{1}{2}x + 3$$

so $\frac{5}{2}x = 3$, or $x = \frac{6}{5}$. This looks plausible from the sketch. We then get $y = \frac{1}{2}x + 3 = \frac{18}{5}$ (and we can check that the equation of l gives the same y -value). Thus the area of the triangle is

$$\frac{1}{2} \times 9 \times \frac{18}{5} = \frac{81}{5} = 16\frac{1}{5}$$

and the correct answer is A.

Question 4

We are dividing a polynomial by a linear expression and asking for the remainder, so it would make sense to use the remainder theorem: if the polynomial $f(x)$ is divided by $x - a$, then the remainder is $f(a)$.

In this case, we start with $(3x^2 + 8x - 3)$ and multiply it by $(px - 1)$. This gives our polynomial $f(x) = (3x^2 + 8x - 3)(px - 1)$. We then divide $f(x)$ by $(x + 1)$ to get a remainder of $f(-1) = 24$ (using the remainder theorem).

Substituting $x = -1$ into $f(x)$ gives

$$f(-1) = (3 - 8 - 3)(-p - 1) = 8(p + 1)$$

Thus $8(p + 1) = 24$, giving $p = 2$, and the correct answer is B.

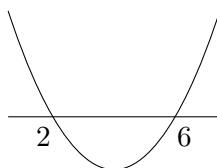
Commentary: We might have been tempted to multiply out the brackets in $f(x)$. But this does not change the value of $f(-1)$, and the remainder theorem does not say anything about the coefficients of x , x^2 and so on, so there is no benefit to be gained from expanding $f(x)$. It would only take more effort and increase the chances of making a mistake.

Question 5

Commentary: The start of this seems like a fairly standard question: find the values of x which satisfy a pair of inequalities. The options offered, though, seem far from standard, so we will have to think carefully once we have found the set S .

We start by solving the two inequalities separately.

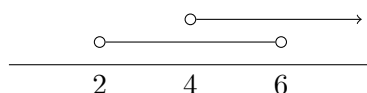
The quadratic factorises as $x^2 - 8x + 12 = (x - 2)(x - 6)$, so we are solving $(x - 2)(x - 6) < 0$. We can sketch the graph of $y = (x - 2)(x - 6)$:



so we see that the first inequality is satisfied when $2 < x < 6$

The second inequality is a linear inequality which immediately simplifies to $2x > 8$ or $x > 4$

We can combine these two inequalities using a number line:



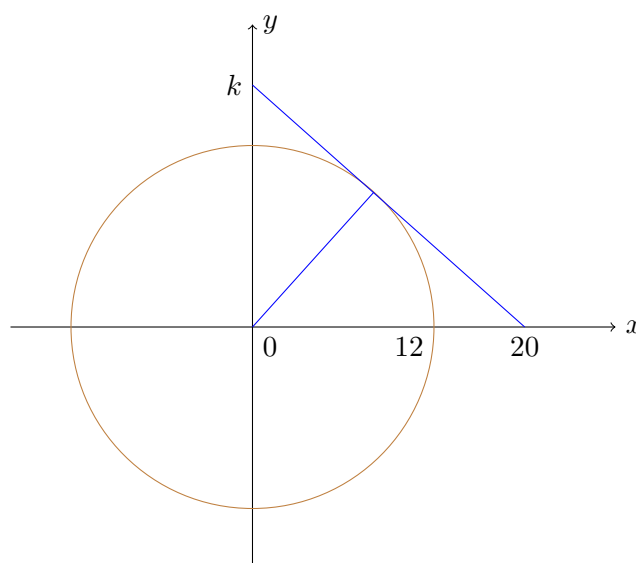
So the set S is given by $4 < x < 6$

Our problem now is that none of the given options says $4 < x < 6$. Most of them are in the form of quadratic inequalities. So we assume that one of the quadratic inequalities probably gives $4 < x < 6$ when we 'solve' it. That, though, seems like a lot of work. We would do better to work in reverse and try to *derive* the quadratic inequality from $4 < x < 6$.

We could think about our earlier sketch, which helped us solve $(x - 2)(x - 6) < 0$ and gave $2 < x < 6$. If we replace the 2 by a 4, we see that the inequality $(x - 4)(x - 6) < 0$ will give $4 < x < 6$. We can expand this to get $x^2 - 10x + 24 < 0$, and so the correct answer is C.

Question 6

This calls for a sketch. The circle $x^2 + y^2 = 144$ is centred on the origin and has radius 12:



It's not exact, and it's not drawn to scale, but it's good enough to work with. The blue radius is perpendicular to the tangent line. We are trying to find the value of k .

Approach 1

We could work out the equation of the tangent line. It has gradient m (which we don't know) and passes through $(20,0)$, so it has equation $y = m(x - 20)$. Since it is tangent to the circle, solving the two equations simultaneously must give one repeated root.

We have $x^2 + y^2 = 144$ into which we can substitute $y = m(x - 20)$ giving

$$\begin{aligned} x^2 + m^2(x - 20)^2 &= 144 \\ \iff x^2 + m^2(x^2 - 40x + 400) &= 144 \\ \iff (m^2 + 1)x^2 - 40m^2x + 400m^2 - 144 &= 0 \end{aligned}$$

We want this to have exactly one root for x , so we need the discriminant to equal zero, giving

$$(-40m^2)^2 - 4(m^2 + 1)(400m^2 - 144) = 0.$$

We can take out a factor of 4^2 :

$$(10m^2)^2 - (m^2 + 1)(100m^2 - 36) = 0$$

and then expand the brackets to give

$$100m^4 - 100m^4 + 36m^2 - 100m^2 + 36 = 0$$

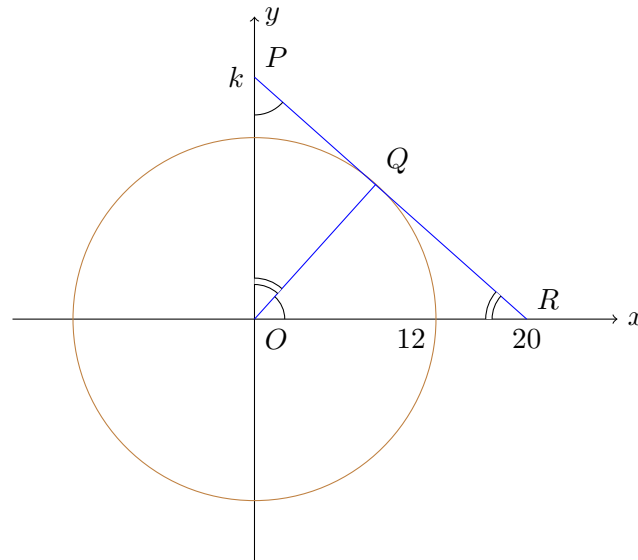
so $64m^2 = 36$, or $8m = \pm 6$. Since $m < 0$ (as the line intersects the positive y -axis), we have $m = -\frac{6}{8} = -\frac{3}{4}$. Thus the tangent line has equation

$$y = -\frac{3}{4}(x - 20)$$

Substituting in $(x, y) = (0, k)$ gives $k = 15$, and hence the answer is B.

Approach 2

We might notice from the sketch that we have some similar triangles, since the radius meets the tangent at right angles. If we label all of the vertices and equal angles, we will be able to describe things more precisely:



So the triangles $\triangle OQR$, $\triangle PQO$ and $\triangle POR$ are all similar. We also know the lengths $OR = 20$ and $OQ = 12$ (as it is a radius). By Pythagoras's Theorem, applied to $\triangle OQR$, we can calculate $QR = 16$ (and we might note that $\triangle OQR$ is a 3-4-5 triangle). We can then use the similarity of triangles $\triangle PQO$ and $\triangle OQR$ to find k :

$$\frac{PO}{QO} = \frac{OR}{QR} \implies \frac{k}{12} = \frac{20}{16}$$

giving $k = 15$, as before.

Commentary: You might prefer one or other of these approaches. The second is shorter, but requires some confidence with geometry and recognising similar triangles.

Question 7

Let the common difference of the arithmetic progression be d and the common ratio of the geometric progression be r . Then we can write the first three terms of each as

$$p, q = p + d, p^2 = p + 2d \quad \text{and} \quad p, p^2 = pr, q = pr^2.$$

The second term of the geometric progression shows that $r = p$ (as p is non-zero), so the third term gives $q = p^3$

Looking again at the arithmetic progression, using $q = p^3$ now shows that the second and third terms are

$$\begin{aligned} p^3 &= p + d \\ p^2 &= p + 2d \end{aligned}$$

We can eliminate d from these by subtracting twice the first equation from the second, giving $p^2 - 2p^3 = -p$. This rearranges to

$$2p^3 - p^2 - p = 0$$

This factorises as $p(2p^2 - p - 1) = 0$, and then to $p(2p + 1)(p - 1) = 0$. This has the three solutions $p = 0$, $p = -\frac{1}{2}$ and $p = 1$. We were told that $p < 0$, so we must have $p = -\frac{1}{2}$.

Thus $d = p^3 - p = -\frac{1}{8} - (-\frac{1}{2}) = \frac{3}{8}$ and the sum of the first 10 terms of the arithmetic progression is given, using the formula $S_n = \frac{1}{2}n(2a + (n - 1)d)$, by

$$S_{10} = \frac{1}{2} \times 10 \left(2 \times \left(-\frac{1}{2} \right) + 9 \times \frac{3}{8} \right) = 5 \left(-1 + \frac{27}{8} \right) = \frac{95}{8}$$

and the correct answer is B.

Question 8

Multiplying out the brackets does not look like a particularly useful approach (what would we do with the terms $\cos x$ and $\sin x \cos x$ in the same inequality?), so we will leave the expression in its factored form.

If we temporarily write $a = 1 - 2\sin x$ and $b = \cos x$ we will be able to see the structure of the problem a little more clearly. We want $ab \geq 0$, and this will happen when either $a \geq 0$ and $b \geq 0$, or when $a \leq 0$ and $b \leq 0$. So we need to solve each of these simpler inequalities.

We have $1 - 2\sin x = 0$ when $\sin x = \frac{1}{2}$, so when $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$. We can either sketch the graph of $y = 1 - 2\sin x$ or try values to determine whether $1 - 2\sin x$ is positive or negative in each range. Sketching the graph seems possibly hard, as there are multiple transformations, and it would be easy to make a mistake. So on this occasion, we'll plug in values: $x = 0$ and $x = \pi$ both give $1 - 2\sin x = 1$; while $x = \frac{\pi}{2}$ gives $1 - 2\sin x = -1$. Therefore

$$\begin{aligned} 1 - 2\sin x &\geq 0 \text{ when } 0 \leq x \leq \frac{\pi}{6} \text{ and when } \frac{5\pi}{6} \leq x \leq \pi \\ 1 - 2\sin x &\leq 0 \text{ when } \frac{\pi}{6} \leq x \leq \frac{5\pi}{6} \end{aligned}$$

We can do similarly for $\cos x$; this time we can think about the graph in our head, and we have

$$\begin{aligned} \cos x &\geq 0 \text{ when } 0 \leq x \leq \frac{\pi}{2} \\ \cos x &\leq 0 \text{ when } \frac{\pi}{2} \leq x \leq \pi \end{aligned}$$

We can now put everything together. Both expressions are greater than or equal to 0 when $0 \leq x \leq \frac{\pi}{6}$, and they are both less than or equal to 0 when $\frac{\pi}{2} \leq x \leq \frac{5\pi}{6}$. Hence the answer is A.

Question 9

We clearly need to work out the radius of the circle to get anywhere on this problem. We can do this by completing the square. The equation of the circle, $x^2 + y^2 - 18x - 22y + 178 = 0$, becomes

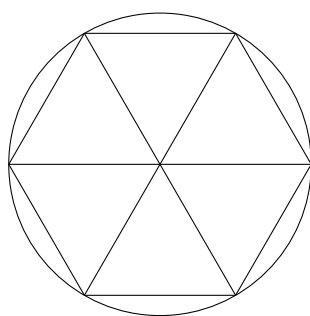
$$(x - 9)^2 - 9^2 + (y - 11)^2 - 11^2 + 178 = 0$$

With a little bit of arithmetic, we can rearrange this to give

$$(x - 9)^2 + (y - 11)^2 = 24$$

so the circle has a radius of $r = \sqrt{24}$

We now need to work out the area of the inscribed regular hexagon. We sketch the hexagon and its diameters:

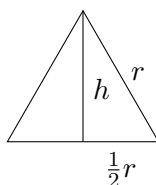


It is clear from the sketch (and even from a not-very-precise sketch) that the hexagon is composed of six triangles, each of which is equilateral and has side length equal to the circle's radius r .

We can work out the area of one of the triangles either using Pythagoras's Theorem or using the formula $\frac{1}{2}ab \sin C$ for the area of a triangle.

The trigonometric formula $\frac{1}{2}ab \sin C$ gives $\frac{1}{2}r^2 \sin 60^\circ = \frac{\sqrt{3}}{4}r^2$. Substituting $r^2 = 24$ then gives each triangle an area of $6\sqrt{3}$, and so the whole hexagon has an area of $36\sqrt{3}$, which is option F.

Alternatively we can proceed using Pythagoras's Theorem. If the height of the triangle is h , then we have:



Therefore $h^2 = r^2 - (\frac{1}{2}r)^2$ giving $h = \frac{\sqrt{3}}{2}r$ and the triangle has area $\frac{1}{2}rh = \frac{\sqrt{3}}{4}r^2$ as before.

Question 10

The gradient of the tangent at x is $f'(x) = -6p^2 + 6px - 3x^2$. Note that p is a constant, and so p^3 differentiates to 0.

The gradient of the tangent where $x = -1$ is therefore $-6p^2 - 6p - 3$, and so the gradient of the normal is the negative reciprocal of this, namely

$$M = \frac{-1}{-6p^2 - 6p - 3} = \frac{1}{3(2p^2 + 2p + 1)}$$

The greatest possible value of this occurs when the denominator of this fraction is as small as possible and positive. We can complete the square to find this:

$$2p^2 + 2p + 1 = 2\left(p + \frac{1}{2}\right)^2 + \frac{1}{2}$$

so the minimum of this is $\frac{1}{2}$ (occurring when $p = -\frac{1}{2}$).

We could also have found this using calculus: $\frac{d}{dp}(2p^2 + 2p + 1) = 4p + 2$ which is zero when $p = -\frac{1}{2}$, giving a minimum value of $\frac{1}{2}$.

Therefore the greatest possible value of M is $\frac{1}{3 \times \frac{1}{2}} = \frac{2}{3}$ which is option E.

Commentary: Note that, in this case, the denominator of the fraction expression for M is always positive. If it was instead something like $p^2 - 1$, then there would not be a greatest possible value of M , as $p^2 - 1$ could get as close to zero as we like, so M could be as large as we like.

Question 11

We have $x_1 = 7$, $x_2 = 3$, $x_3 = 1$. It is not at all clear what the pattern here is, so we calculate the next few terms:

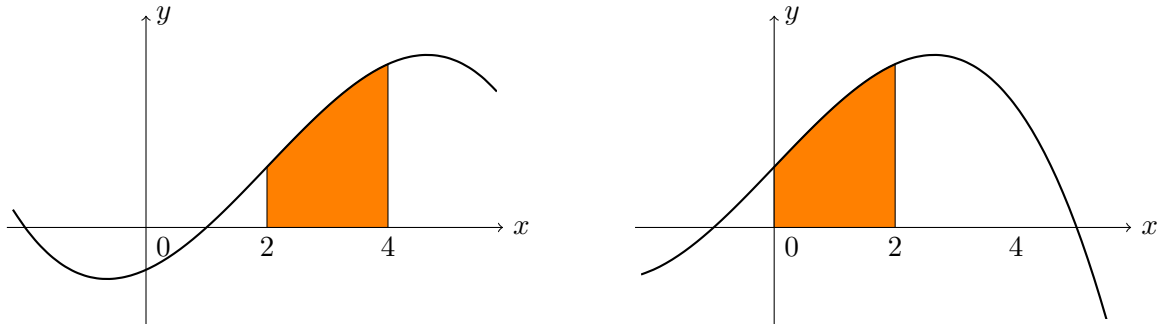
$$x_4 = \frac{23 \times 1 - 53}{5 \times 1 + 1} = \frac{-30}{6} = -5$$
$$x_5 = \frac{23 \times (-5) - 53}{5 \times (-5) + 1} = \frac{-168}{-24} = 7$$

so $x_6 = 3$, $x_7 = 1$, $x_8 = -5$ and so on. Thus the terms repeat in a cycle with length 4, and as $100 = 25 \times 4$, we see that $x_{100} = -5$, which is option A.

Question 12

We have $\int_2^4 f(x) dx = A$, and the question asks about $\int f(x+2) dx$ between various limits.

The graph of $y = f(x+2)$ is the graph of $y = f(x)$ translated left by 2 units, so we can draw a sketch to see what effect this has on the area between $x = 2$ and $x = 4$:



This means that $\int_2^4 f(x) dx = \int_0^2 f(x+2) dx = A$

(If you have learnt about integration by substitution, you can also show this as follows: writing $x = u + 2$ gives $dx = du$, and $u = x - 2$, so $\int_2^4 f(x) dx = \int_0^2 f(u+2) du$)

Therefore the answer is either A or B. We have

$$\int_0^2 [f(x+2) + 1] dx = \int_0^2 f(x+2) dx + \int_0^2 1 dx$$

The first integral equals A , as we have shown, and $\int_0^2 1 dx = 2$, so the whole integral equals $A + 2$ and the correct answer is B.

Question 13

The term in x^4 is $\binom{5}{4}a(bx)^4 = 5ab^4x^4$, so the coefficient of x^4 is $5ab^4$

The term in x^2 is $\binom{5}{2}a^3(bx)^2 = 10a^3b^2x^2$, so the coefficient of x^2 is $10a^3b^2$

It follows, since the coefficient of x^4 is 8 times that of x^2 , that

$$5ab^4 = 80a^3b^2$$

As a and b are non-zero, we can divide by $5ab^2$ to get $b^2 = 16a^2$. Since both a and b are positive, we can take square roots to get $b = 4a$. (If we didn't know about their signs, we could only say that $b = \pm 4a$.)

So the smallest possible value of $a + b$ occurs when $a = 1$, $b = 4$, giving $a + b = 5$, which is option C.

Question 14

The equations involve 2^x and $2^{2x} = (2^x)^2$, so we write $a = 2^x$; likewise we write $b = 2^y$, and so our equations become:

$$\begin{aligned}a + 3b &= 3 \\ a^2 - 9b^2 &= 6\end{aligned}$$

The first equation gives $a = 3 - 3b$, which we can substitute into the second equation to obtain

$$(3 - 3b)^2 - 9b^2 = 6$$

which expands to $9 - 18b = 6$, and hence $b = \frac{1}{6}$ and $a = \frac{5}{2}$

Another way to solve these simultaneous equations is to note that the second one can be written as $(a + 3b)(a - 3b) = 6$, using the identity for the difference of two squares. Since $a + 3b = 3$, this gives $a - 3b = 2$, from which we can again deduce a and b .

We can now find x and y :

$$\begin{aligned}x &= \log_2 a = \log_2 \frac{5}{2} = p \\ y &= \log_2 b = \log_2 \frac{1}{6} = q\end{aligned}$$

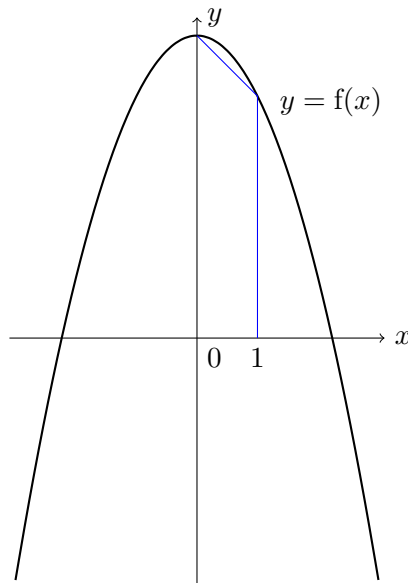
and hence

$$p - q = \log_2 \frac{5}{2} - \log_2 \frac{1}{6} = \log_2 \left(\frac{5}{2} \div \frac{1}{6} \right) = \log_2 15$$

Thus the correct answer is F.

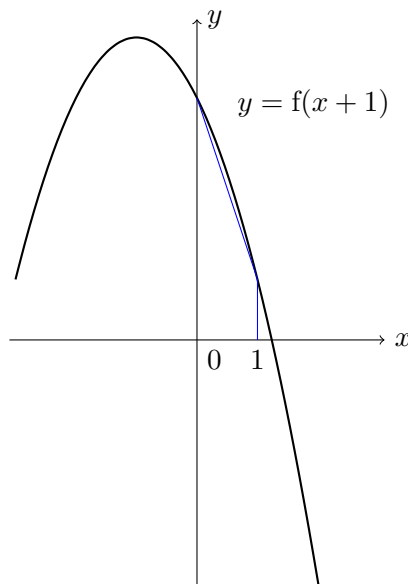
Question 15

We can either proceed by sketching the graph or by using a general theorem about the trapezium rule. We start with sketching. The simplest case of the trapezium rule is where we only have one trapezium, so that is what we will draw.



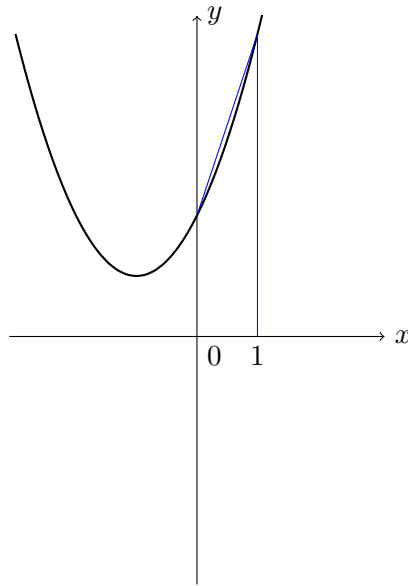
We see here that the trapezium rule underestimates the area of $y = f(x)$

For the case $y = f(x + 1)$ the graph of $y = f(x)$ is translated one unit to the left:



Again, the trapezium rule gives an underestimate.

For the third case, we need to reflect this graph in the line $y = 6$, which gives the following graph:



We see that this time, the trapezium rule gives an overestimate.

So the correct answer is B.

An alternative approach is given by noting that the trapezium rule gives an overestimate of the area under the graph of $y = f(x)$ if $f''(x) > 0$ throughout the range and an underestimate if $f''(x) < 0$ throughout. In our case, $f''(x) = -4$, so the trapezium rule gives an underestimate for (1). For case (2), $f''(x + 1) = -4$ again, so we still have an underestimate. For the third case, the equation of the graph is $y = 12 - f(x + 1)$, so the second derivative is $-f''(x + 1) = 4$, and therefore we have an overestimate.

Question 16

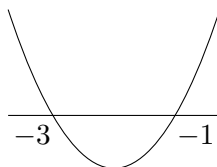
In this question, we consider a function to be increasing when $f'(x) \geq 0$, and decreasing when $f'(x) \leq 0$.

Commentary: The classification of increasing and decreasing we use in the options for this question avoids a subtle issue: what we consider a function to be doing (i.e., increasing, decreasing, or neither) at points where $f'(x) = 0$. For this question, though, we don't need to concern ourselves with this as the options all use $\geq$ or $\leq$. However, we will set out our answer using ' $\geq$ ' or ' $\leq$ '.

We therefore calculate $f'(x)$ and $g'(x)$ and find where each is positive and negative.

$f'(x) = 6x + 12 = 6(x + 2)$, so $f'(x) > 0$ when $x > -2$ and $f'(x) < 0$ when $x < -2$.

$g'(x) = 3x^2 + 12x + 9 = 3(x^2 + 4x + 3)$. We can find where this is positive and negative by sketching the graph: $x^2 + 4x + 3 = (x + 1)(x + 3)$, giving the sketch of $g'(x)$:



so $g'(x) > 0$ when $x < -3$ or $x > -1$, and $g'(x) < 0$ when $-3 < x < -1$.

We now want one of $f'(x)$ and $g'(x)$ to be positive and the other negative.

We have $f'(x) > 0$ and $g'(x) < 0$ when $x > -2$ and $-3 < x < -1$, so when $-2 < x < -1$.

Similarly, $f'(x) < 0$ and $g'(x) > 0$ when $x < -2$ and either $x < -3$ or $x > -1$; together these give $x < -3$.

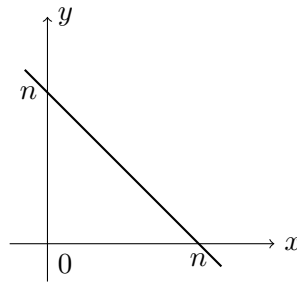
So the complete set of solutions (if we decide to include the points where $f'(x) = 0$) is $x \leq -3$ or $-2 \leq x \leq -1$, and so the answer is option E.

Question 17

It seems that we need to calculate $F(n)$ before we can find $G(n)$. We could do this either using integration or geometry. We start with an integration method:

$$\begin{aligned} F(n) &= \frac{1}{n} \int_0^n (n-x) \, dx \\ &= \frac{1}{n} \left[nx - \frac{1}{2}x^2 \right]_0^n \\ &= \frac{1}{n} \left(n^2 - \frac{1}{2}n^2 \right) \\ &= \frac{1}{2}n \end{aligned}$$

We could reach the same conclusion by sketching a graph of the function:



The area of this triangle is $\frac{1}{2}n^2$, so $F(n) = \frac{1}{n} \times \frac{1}{2}n^2 = \frac{1}{2}n$

We now have:

$$\begin{aligned} G(n) &= \sum_{r=1}^n F(r) \\ &= \sum_{r=1}^n \frac{1}{2}r \\ &= \frac{1}{2} \sum_{r=1}^n r \end{aligned}$$

where for the last line we have used $(\frac{1}{2}F(1) + \frac{1}{2}F(2) + \dots) = \frac{1}{2}(F(1) + F(2) + \dots)$

Now $\sum_{r=1}^n r = 1 + 2 + \dots + n = \frac{1}{2}n(n+1)$, either using the formula for triangular numbers or of the sum of an arithmetic series.

So $G(n) = \frac{1}{4}n(n+1)$.

Therefore

$$\begin{aligned} G(n) &> 150 \\ \iff \frac{1}{4}n(n+1) &> 150 \\ \iff n(n+1) &> 600 \end{aligned}$$

So we can try values: 25 is a nice number to calculate with, so we start with options C and D:

$$24 \times 25 = 600$$

$$25 \times 26 = 650$$

so $n = 25$ is the smallest positive integer for which $G(n) > 150$, and the answer is D.

Question 18

We can do this algebraically.

A translation of the graph of $y = f(x)$ by 2 units in the positive y -direction gives the graph of $y = f(x) + 2$

A stretch of factor k parallel to the x -axis gives the graph of $y = f(\frac{1}{k}x) = f(\frac{x}{k})$

For these to be the same, we need $f(x) + 2 = f(\frac{1}{k}x)$ for all (positive) x .

In our case, $f(x) = \log_{10} x$, so we require

$$\log_{10} x + 2 = \log_{10}(\frac{x}{k})$$

But $\log_{10}(\frac{x}{k}) = \log_{10} x - \log_{10} k$, so we require $\log_{10} k = -2$, giving $k = 10^{-2} = 0.01$. Hence the answer is A.

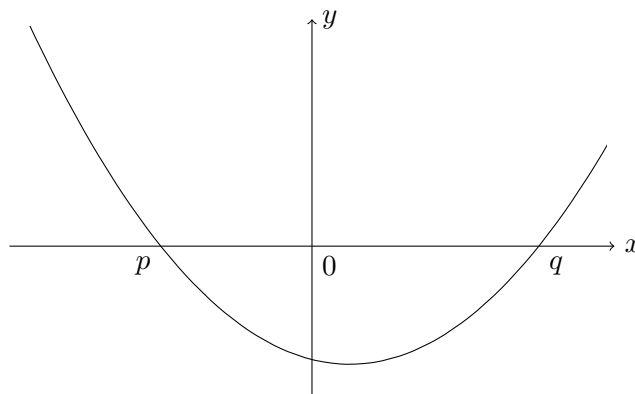
We have done more than required to answer this question: we have actually shown that these transformations are equivalent. An alternative approach is to assume that the question is correct and that these transformations are equivalent, and to use a single point to find the value of k .

We could see which point ends up on the x -axis after the translation: it was at $(a, -2)$, so it must have been at $(0.01, -2)$. Therefore the translated graph crosses the x -axis at 0.01, while the original graph crossed the x -axis at 1. Therefore the stretch needs to be by a factor of 0.01 parallel to the x -axis.

We could have done a similar calculation with any other point.

Question 19

We can sketch the graph of $y = x^2 + bx + c$. It intersects the x -axis at p and q , and intersects the y -axis at $c < 0$. So the graph must look like this:



However, the vertex might be either side of the y -axis.

To solve the inequality $x^2 + bcx + c^3 < 0$, we need to find where this equals zero. We can use the quadratic formula for this: $x^2 + bcx + c^3 = 0$ when

$$\begin{aligned} x &= \frac{-bc \pm \sqrt{(bc)^2 - 4c^3}}{2} \\ &= \frac{1}{2}(-bc \pm \sqrt{b^2c^2 - 4c^3}) \\ &= \frac{1}{2}c(-b \pm \sqrt{b^2 - 4c}) \end{aligned}$$

Going back to the original quadratic, we can solve it to find p and q in terms of b and c :

$$p, q = \frac{1}{2}(-b \pm \sqrt{b^2 - 4c})$$

Comparing these, we see that the roots of $x^2 + bcx + c^3$ are pc and qc in some order. So the solution is C or D.

To distinguish between these possibilities, we need to know which is the larger of pc and qc . Note that we were told that $c < 0$. Since $p < q$, it follows that $pc > qc$, and so the answer is D.

It turns out that we didn't actually need our graph sketch.

A somewhat less obvious approach to this question is as follows.

We can divide the second inequality by $c^2 > 0$, so $x^2 + bcx + c^3 < 0$ is equivalent to

$$\frac{x^2}{c^2} + \frac{bx}{c} + c < 0$$

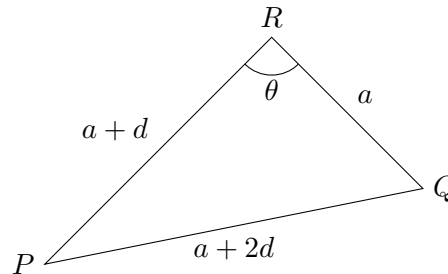
or

$$\left(\frac{x}{c}\right)^2 + b\left(\frac{x}{c}\right) + c < 0$$

This is the same as our original inequality, just with x replaced by $\frac{x}{c}$, so its solution is $p < \frac{x}{c} < q$. Multiplying this by $c < 0$ then gives the solution $pc > x > qc$.

Question 20

It would be really helpful to sketch this triangle so that we don't get confused! We will write θ for the angle $\angle PRQ$.



It is clear that we want to use the cosine rule to relate the three sides and the angle θ . Using ' $c^2 = a^2 + b^2 - 2ab \cos C$ ', we have

$$(a + 2d)^2 = a^2 + (a + d)^2 - 2a(a + d) \cos \theta$$

which we can expand and simplify to give

$$a^2 + 4ad + 4d^2 = 2a^2 + 2ad + d^2 - 2a(a + d) \cos \theta$$

Rearranging gives

$$\cos \theta = \frac{a^2 - 2ad - 3d^2}{2a(a + d)}$$

We can simplify this by noticing that the numerator factorises:

$$\cos \theta = \frac{(a + d)(a - 3d)}{2a(a + d)} = \frac{a - 3d}{2a}$$

Since $3d > 2a$, $a - 3d < -a$, and so $\cos \theta < \frac{-a}{2a} = -\frac{1}{2}$. This gives $\theta > 120^\circ$, and so the answer must be option E.

We can go further and show that $\theta < 180^\circ$. Since PQR is a triangle, the length PQ must be less than the sum of PR and RQ , that is $a + 2d < a + d + a$, so $d < a$. This gives $\cos \theta > \frac{-2a}{2a} = -1$ so $\theta < 180^\circ$.

An alternative approach is to just try values for d . We are given that $d > \frac{2}{3}a$, so we could try $d = \frac{2}{3}a$. This gives side lengths of $a, \frac{5}{3}a, \frac{7}{3}a$, which we can take to be 3, 5, 7 if we choose $a = 3$. Plugging these lengths into the cosine rule gives $\theta = 120^\circ$, so this is one limit.

We are allowed larger values of d . If we take d too large, then we will no longer have a triangle, as $PQ > PR + RQ$. So as d increases, the angle θ must increase too. Therefore the answer must be E: $120^\circ < \theta < 180^\circ$.

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Test of Mathematics for University Admission

Paper 2 2017 worked answers

Test of Mathematics for University Admission

2017 Paper 2

Worked Solutions

Version 1.0, April 2019

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Introduction for students

These solutions are designed to support you as you prepare to take the Test of Mathematics for University Admission. They are intended to help you understand how to answer the questions, and therefore you are strongly encouraged to **attempt the questions first** before looking at these worked solutions. For this reason, each solution starts on a new page, so that you can avoid looking ahead.

The solutions contain much more detail and explanation than you would need to write in the test itself – after all, the test is multiple choice, so no written solutions are needed, and you may be very fluent at some of the steps spelled out here. Nevertheless, doing too much in your head might lead to making unnecessary mistakes, so a healthy balance is a good target!

There may be alternative ways to correctly answer these questions; these are not meant to be ‘definitive’ solutions.

The questions themselves are available on the ‘**Preparing for the test**’ section on the Admissions Testing website.

Paper 2 uses ideas from mathematical logic. These are explained in detail in the ‘Notes on Logic and Proof’ on the above webpage.

Question 1

We first expand the brackets and split up the fraction:

$$\begin{aligned} y &= \frac{(1-3x)^2}{2x^{\frac{3}{2}}} \\ &= \frac{1-6x+9x^2}{2x^{\frac{3}{2}}} \\ &= \frac{1}{2}x^{-\frac{3}{2}} - 3x^{-\frac{1}{2}} + \frac{9}{2}x^{\frac{1}{2}} \end{aligned}$$

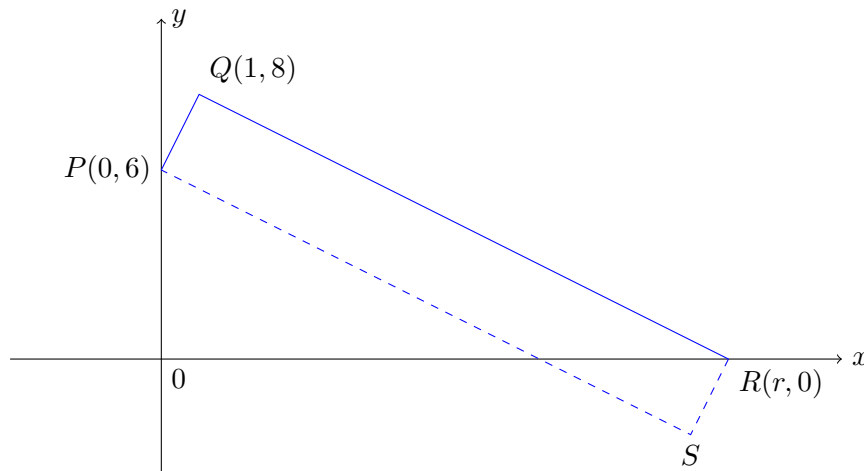
and we can now differentiate this to get

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2}\left(-\frac{3}{2}\right)x^{-\frac{5}{2}} - 3\left(-\frac{1}{2}\right)x^{-\frac{3}{2}} + \frac{9}{2} \times \frac{1}{2}x^{-\frac{1}{2}} \\ &= -\frac{3}{4}x^{-\frac{5}{2}} + \frac{3}{2}x^{-\frac{3}{2}} + \frac{9}{4}x^{-\frac{1}{2}} \end{aligned}$$

and so the correct answer is A.

Question 2

A quick sketch (not drawn to scale) may help us here.



PQ has length $\sqrt{1^2 + 2^2} = \sqrt{5}$ by Pythagoras's theorem.

The gradient of PQ is $\frac{8-6}{1-0} = 2$, so QR has gradient $-\frac{1}{2}$

We could now find the coordinates of R by finding the equation of the line QR and substituting in $y = 0$ to find the x -coordinate of R .

Alternatively, if we set the coordinates of R to be $(r, 0)$, we can just use the known gradient of QR to find r :

$$\frac{0-8}{r-1} = -\frac{1}{2}$$

so $\frac{8}{r-1} = \frac{1}{2}$ giving $r-1 = 16$, hence $r = 17$

Therefore the length QR is

$$\sqrt{(r-1)^2 + (-8)^2} = \sqrt{16^2 + 8^2} = 8\sqrt{2^2 + 1^2} = 8\sqrt{5}$$

which gives the area of $PQRS$ as $\sqrt{5} \times 8\sqrt{5} = 40$, and the answer is E.

Interestingly, it turns out that we only actually needed $r-1$; we did not need to know the actual coordinates of R .

Question 3

The first term is $a = 2\sqrt{3}$ and the common ratio is r . So the fourth term is $ar^3 = 2\sqrt{3}r^3 = \frac{9}{4}$ and hence

$$r^3 = \frac{9}{8\sqrt{3}} = \frac{9\sqrt{3}}{8 \times 3} = \frac{3\sqrt{3}}{8}$$

by rationalising the denominator.

We can write $3\sqrt{3}$ as $\sqrt{27}$ or $3^{\frac{3}{2}}$ (since we want to take the cube root of this expression), and so we have

$$r^3 = \frac{3^{\frac{3}{2}}}{8} \implies r = \frac{3^{\frac{1}{2}}}{2} = \frac{\sqrt{3}}{2}.$$

(There is a unique real cube root of any real number, unlike the case for square roots.)

We can now find the sum to infinity of the geometric progression:

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ S_{\infty} &= \frac{2\sqrt{3}}{1 - \frac{\sqrt{3}}{2}} \\ &= \frac{4\sqrt{3}}{2 - \sqrt{3}} && \text{multiplying by } \frac{2}{2} \\ &= \frac{4\sqrt{3}(2 + \sqrt{3})}{(2 - \sqrt{3})(2 + \sqrt{3})} && \text{rationalising the denominator} \\ &= \frac{4(2\sqrt{3} + 3)}{1} && \text{paying attention to the options offered} \\ &= 4(2\sqrt{3} + 3) \end{aligned}$$

and the correct answer is G.

Question 4

Let's run through every step of the argument with care.

The first step is ' $\tan x = \sqrt{3}$ so $x = 60^\circ$ '. We should first check that $x = 60^\circ$ is a solution: using our knowledge of trigonometry and special angles, we know that $\tan 60^\circ = \sqrt{3}$, so it is.

But we also know that $\tan$ is periodic with period 180° : there are multiple values of x which have $\tan x = \sqrt{3}$, and all solutions to $\tan x = \sqrt{3}$ are therefore of the form $x = 60^\circ + 180n^\circ$ for $n \in \mathbb{Z}$.

So there is unquestionably an error in the student's answer, but we do not yet know whether there is more than one possible value of $\sin 2x$.

It follows from $x = 60^\circ + 180n^\circ$ that $2x = 120^\circ + 360n^\circ$

It is certainly the case that $\sin 120^\circ = \frac{\sqrt{3}}{2}$. But since $\sin$ is periodic with period 360° , we see that $\sin(120^\circ + 360n^\circ) = \frac{\sqrt{3}}{2}$ for every $n \in \mathbb{Z}$.

So there is only one possible value of $\sin 2x$, but the student should have considered other possible values of x for which $\tan x = \sqrt{3}$; this is option B.

Question 5

We note from the context of the statements that $91 = 7 \times 13$ shows that 91 is not prime. It also shows that 91 is a multiple of 7 and a multiple of 13.

We use a counterexample when we are trying to disprove a statement of the form ‘**if** A **then** B ’; the counterexample will be a case where A is true but B is false. We can also use counterexamples to disprove statements of the form ‘**for all** x , A is true’: a single example of an x for which A is false shows that the statement is false.

We first rewrite each of the statements given into an explicit ‘**if** ... **then**’ statement or an explicit ‘**for all**’ statement.

(With experience, this is not actually necessary; once you have seen enough statements, it becomes clear what a counterexample ‘looks like’. But for the purposes of reaching this point, it can be useful to go through these formal steps.)

- 1 The word ‘when’ in this context has the same logical meaning as ‘if’, so the statement can be rewritten as ‘**if** p is an odd prime, **then** $10p^2 + 1$ and $10p^2 - 1$ are both prime’.

So if $91 = 7 \times 13$ is to provide a counterexample, we need an odd prime p with $10p^2 + 1 = 91$ or $10p^2 - 1 = 91$, since in that case, ‘ $10p^2 + 1$ and $10p^2 - 1$ are both prime’ will be false. It is clear that $p = 3$ achieves this, for then $10p^2 + 1 = 91$.

- 2 We can write this statement as an explicit ‘**for all**’ statement, as ‘Every prime ...’ suggests this meaning. The statement becomes: ‘**For all** primes p where $p > 5$, $p = 6n + 1$ for some integer n ’.

So a counterexample would be a prime p with $p > 5$ but for which $p = 6n + 1$ is false. However, 91 is not prime, so $91 = 7 \times 13$ does not provide a counterexample. The fact that $91 = 6n + 1$ for $n = 15$ is immaterial, and the fact that the statement is false (as $11 = 6 \times 2 - 1$) does not help either.

- 3 We can again write this as an explicit ‘**for all**’ statement; the English is better with the equivalent ‘**for each**’, though: ‘**For each** n which is a multiple of 7 greater than 7, n is not prime.’

If we now look at $91 = 7 \times 13$, we see that 91 is a multiple of 7 greater than 7. However, 91 is not prime, so 91 is an example of where the statement does hold, rather than being a counterexample. In fact, the given statement is true, so there cannot be any counterexamples, and we need not have looked at $91 = 7 \times 13$ at all.

So the answer is B.

Question 6

We calculate the first few values of u_n in the hope that this will show a clear pattern.

We have

$$\begin{aligned}
 u_0 &= 1 \\
 u_1 &= \int_0^1 4x.1 \, dx \\
 &= [2x^2]_0^1 \\
 &= 2 \\
 u_2 &= \int_0^1 4x.2 \, dx \\
 &= [4x^2]_0^1 \\
 &= 4 \\
 u_3 &= \int_0^1 4x.4 \, dx \\
 &= [8x^2]_0^1 \\
 &= 8
 \end{aligned}$$

The pattern is now apparent: $u_n = 2^n$, and so the answer is A.

If we wanted to *prove* this to be true, we could use induction. The result is clearly true when $n = 0$, so assume that it is true when $n = k$. Then

$$\begin{aligned}
 u_{k+1} &= \int_0^1 4x.2^k \, dx \\
 &= [2.2^k x^2]_0^1 \\
 &= 2.2^k \\
 &= 2^{k+1}
 \end{aligned}$$

and so the result holds for $n = k + 1$. Hence by induction, $u_n = 2^n$ for all $n \geq 0$.

Question 7

The graph of $y = f(x)$ looks like an exponential graph, so all of the four given statements certainly seem plausible.

One approach to this question is to think about the shapes of different exponential graphs. Another is to consider a specific value of x , and see what happens.

If we take $x = 1$, $a^x = a$, and $f(1) > a$. We can now substitute $x = 1$ into the two expressions for $f(x)$ offered: $f(x) = b^x$ gives $f(1) = b$, so we would need $b > a$. Similarly, $f(x) = a^{kx}$ gives $f(1) = a^k$. But it is clear from the solid line that $a > 1$ (as $a^0 = 1$ and $a^1 = a > 1$), so $a^k > a$ requires $k > 1$.

So both statements **1** and **3** could be true, but not **2** or **4**.

We can be even more careful and show that we cannot have just one of **1** and **3** being true. If statement **3** is true, we have $f(x) = a^{kx} = (a^k)^x$, so writing $b = a^k > a$ shows that statement **1** is true; likewise, if statement **1** is true, we have $f(x) = b^x = (a^{\log_a b})^x = a^{(\log_a b)x}$ and $\log_a b > 1$, so statement **3** is also true.

Hence the correct answer is E.

Question 8

We can start by estimating or calculating the values.

A $2 < \log_2 7 < 3$, since $\log_2 4 = 2$ and $\log_2 8 = 3$

B $(2^{-3} + 2^{-2})^{-1} = (\frac{1}{8} + \frac{1}{4})^{-1} = (\frac{3}{8})^{-1} = \frac{8}{3}$ which again lies between 2 and 3

C $\pi/3$ is just a little greater than 1, so $2^{\pi/3}$ is a little greater than 2

D We expand $(\sqrt{2} - 1)^3$ using the binomial theorem to get

$$(\sqrt{2})^3 - 3(\sqrt{2})^2 + 3\sqrt{2} - 1 = 2\sqrt{2} - 6 + 3\sqrt{2} - 1 = 5\sqrt{2} - 7$$

Thus

$$\frac{1}{4(\sqrt{2} - 1)^3} = \frac{1}{4(5\sqrt{2} - 7)} = \frac{5\sqrt{2} + 7}{4((5\sqrt{2})^2 - 7^2)} = \frac{5\sqrt{2} + 7}{4}$$

This is greater than $\frac{5+7}{4} = 3$, so is certainly not the smallest in value.

E $\sin(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$, so $4\sin^2(\frac{\pi}{4}) = 4(\frac{1}{\sqrt{2}})^2 = 4 \times \frac{1}{2} = 2$

Since A, B and C are each greater than 2, the correct answer is E.

Question 9

Commentary: This is Fermat's Last Theorem for the case $n = 3$; it was first proven by Euler in the 18th century. The attempted proof here, though, is deeply flawed, as we will see.

Line I is fine: this is a simple rearrangement.

Line II is fine: this is a standard algebraic identity which is easy to check.

Line III is problematic. It is certainly true that $a \leq a^2$ and $c - b < c + b \leq c^2 + b^2 < c^2 + cb + b^2$. It is therefore plausible that $a = c - b$ and $a^2 = c^2 + cb + b^2$, but it is far from necessary. Of course we cannot give an explicit example to show this, as there are no sets of positive integers a , b and c with $a^3 + b^3 = c^3$. But we can observe that if $c - b = 1$, we must have $a^3 = c^2 + bc + b^3$, and this possibility has not been considered; alternatively, we could consider a case where a is not prime such as $a = 6$; then we can write $a^3 = 6^3 = 3 \times 72$, and we could have $c - b = 3$ and $c^2 + cb + b^2 = 72$. Either way, it does not necessarily follow that $a = c - b$ and $a^2 = c^2 + cb + b^2$, so this step is wrong.

The correct answer is D.

Question 10

To show that a condition is **sufficient**, we need to show that if the condition is satisfied, then $\int_1^3 f(x) dx = 0$

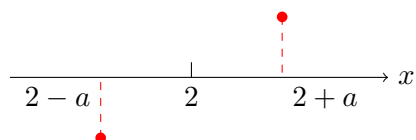
Note that it is not enough for us to find an example of a function which satisfies $\int_1^3 f(x) dx = 0$ and then to check whether the function satisfies the suggested conditions, as this would be checking whether the conditions are **necessary**, not whether they are **sufficient**.

- A** Knowing the value of $f(x)$ at a single value of x cannot hope to force the integral to be 0; consider, for example, the quadratic $f(x) = (x - 2)^2$
- B** Likewise, knowing the value of $f(x)$ at the endpoints is not **sufficient**; consider, for example, the quadratic $f(x) = (x - 1)(x - 3)$
- C** This says that the function is odd: the values of $f(x)$ for negative values of x match those for positive values of x . But it gives no control over the values of $f(x)$ between $x = 1$ and $x = 3$, so it cannot be a **sufficient** condition. For example, $f(x) = x$ satisfies $f(-x) = -f(x)$, but $\int_1^3 x dx > 0$

So it is one of D and E, both of which look complicated. E looks marginally easier to deal with, as $x - 2 = -(2 - x)$. But that means that if we write $y = 2 - x$, condition E becomes $f(-y) = -f(y)$, which is exactly the same as condition C. So condition E is not **sufficient**.

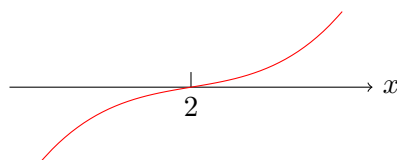
By elimination, therefore, the correct answer must be condition D.

Let us show that D is, indeed, **sufficient**. It would help to see what is going on if we write $x + 2$ as $2 + x$, so the condition becomes $f(2 + x) = -f(2 - x)$. If we take $0 < x \leq 1$, this shows that there is rotational symmetry about the point $(2, 0)$:



And when $x = 0$, we get $f(2) = -f(2)$ so $f(2) = 0$

So the graph of $y = f(x)$ itself might look like this:



Since the graph of $y = f(x)$ has rotational symmetry about $(2, 0)$, the integrals $\int_1^2 f(x) dx$ and $\int_2^3 f(x) dx$ will be negative of each other, and so condition D is **sufficient** for $\int_1^3 f(x) dx = 0$.

Question 11

We can approach this question algebraically or graphically.

We start with an algebraic approach. We first note that $f(x) \geq 0$ for all $x > 0$ as $f(x)$ is increasing. So the region described first, with area R , lies entirely above (or on) the x -axis, and

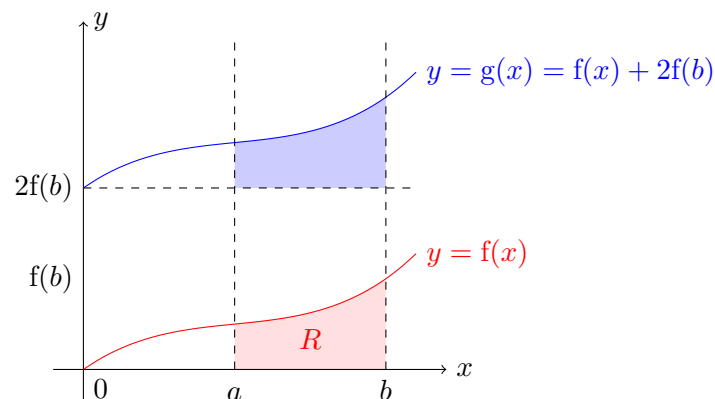
$$R = \int_a^b f(x) \, dx$$

We note that $g(x) = f(x) + 2f(b) \geq 0$ for $x > 0$ as $f(x) \geq 0$ and $f(b) \geq 0$. Then the area required under the graph $g(x)$ is given by

$$\begin{aligned} \int_a^b g(x) \, dx &= \int_a^b f(x) + 2f(b) \, dx \\ &= \int_a^b f(x) \, dx + 2f(b) \int_a^b 1 \, dx \\ &= R + 2f(b)(b - a) \end{aligned}$$

On the second line we have split the integral into a sum of two integrals, and taken out the *constant* factor of $2f(b)$ from the second integral. Thus the answer is option B.

We could also work graphically. We can sketch the graphs of $f(x)$ and $g(x)$, making up a suitable increasing shape for $f(x)$:



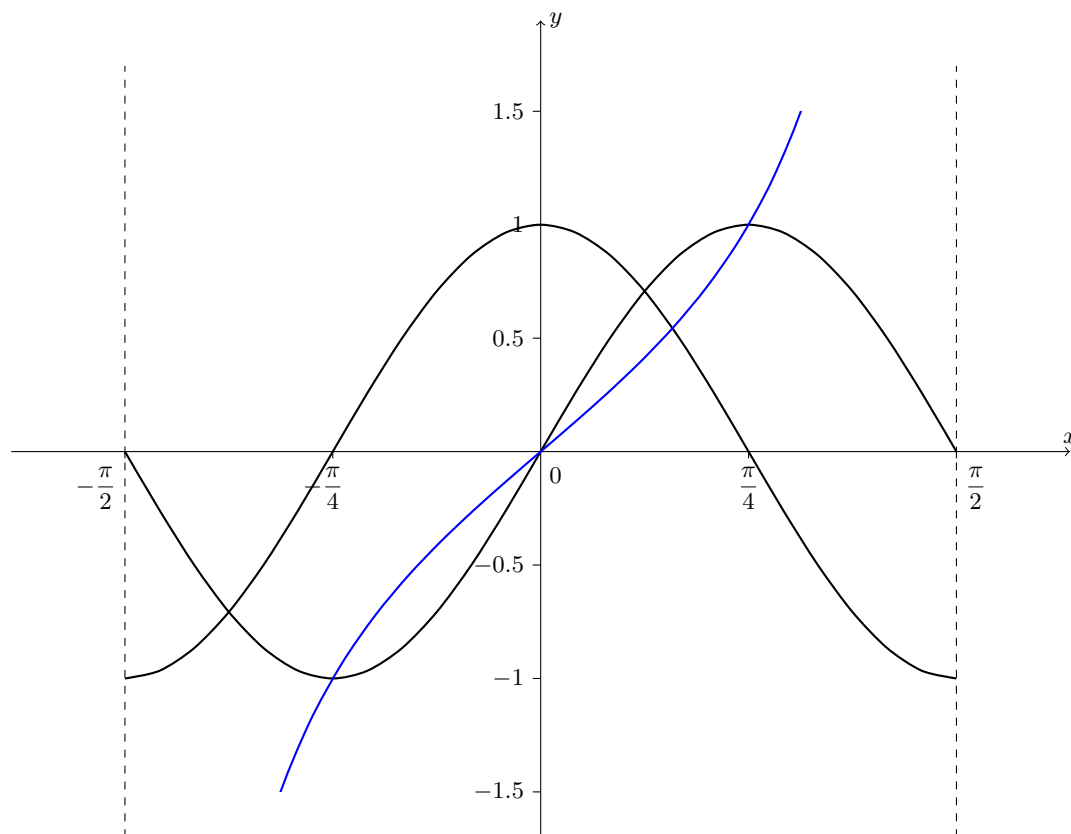
It is clear from this sketch that the area under the graph of $y = g(x)$ is formed of a rectangle which is $(b - a)$ by $2f(b)$ in size, and the area under the original graph, R , so that the required area is $R + (b - a) \times 2f(b)$.

Question 12

We clearly need to sketch the graph of $y = \tan x$ onto the given sketch.

This graph has asymptotes at $x = \pm\frac{\pi}{2}$, and passes through the points $(\frac{\pi}{4}, 1)$ and $(-\frac{\pi}{4}, -1)$. Since the function is increasing as it passes through these points, it must lie below the graph of $y = \sin 2x$ between 0 and $\frac{\pi}{4}$, and above it between $-\frac{\pi}{4}$ and 0.

This results in the following graph:



We now pick various values for x and see which order the functions lie in. For example, when x is close to $-\frac{\pi}{2}$, we have $\tan x < \cos 2x < \sin 2x$. Here are some cases as we move from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$, each together with the option it eliminates:

x value	Function order	Option
x close to $-\frac{\pi}{2}$	$\tan x < \cos 2x < \sin 2x$	F
x just less than $-\frac{\pi}{4}$	$\tan x < \sin 2x < \cos 2x$	E
x just more than $-\frac{\pi}{4}$	$\sin 2x < \tan x < \cos 2x$	D
x just more than 0	$\tan x < \sin 2x < \cos 2x$	E
x just more than $\frac{\pi}{8}$	$\tan x < \cos 2x < \sin 2x$	F
x just less than $\frac{\pi}{4}$	$\cos 2x < \tan x < \sin 2x$	B
x just more than $\frac{\pi}{4}$	$\cos 2x < \sin 2x < \tan x$	A

The only option which is not eliminated is C: $\sin 2x < \cos 2x < \tan x$ for some real number x with $-\frac{\pi}{2} < x < \frac{\pi}{2}$

It is also clear from the graph that this is impossible: the only time that $\tan x$ is greater than both $\sin 2x$ and $\cos 2x$ is when $\frac{\pi}{4} < x < \frac{\pi}{2}$, and in the whole of that range, $\cos 2x < \sin 2x$

Question 13

One way to think about this is to write it as a column addition. If we write $a \times 10^{-3}$ as $0.00a_1a_2\dots$, where $a = a_1.a_2a_3\dots$ (as a decimal expansion), and similarly for the other numbers, the sum becomes:

$$\begin{array}{r} 0 . 0 \ 0 \ a_1 \ a_2 \ \dots \\ + 0 . 0 \ b_1 \ b_2 \ b_3 \ \dots \\ \hline 0 . c_1 \ c_2 \ c_3 \ c_4 \ \dots \end{array}$$

We see that, as $c_1 \geq 1$, there must be a carry from the hundredths column to the tenths column, and hence $b_1 = 9$, so $b \geq 9$. If $b = 9$ exactly, then there cannot be a carry; we would just get a sum of $0.09a_1a_2\dots$ in that case. So we must have $b > 9$, and **II must be true**.

There is no need to have $a_1 = 9$; we could, for example, have $a_1 = 8$ and $b_2 = 3$ and still have a carry. So **I is not necessary true**.

Comparing a and b to c , we know that $1 \leq c < 2$, as $c_1 = 1$. We saw that $a_1 = 8$ is possible, so $a > 8$ is possible, and $a < c$ (**statement III**) is not necessarily true.

We have already shown that $b > 9$, so since $c < 2$, $b > c$ (**statement IV**) is never true.

So the answer is B.

Question 14

We can complete the square for the original quadratic:

$$x^2 - 2bx + c = (x - b)^2 - b^2 + c$$

so the vertex P is at $(b, c - b^2)$.

The graph given shows that $b > 0$ and $c - b^2 > 0$.

The new quadratic is

$$x^2 - 2Bx + c = (x - B)^2 - B^2 + c$$

so the new vertex is at $(B, c - B^2)$.

Now $B > b$, so the new vertex is to the right of P . As $B > b > 0$, $B^2 > b^2$ so $c - B^2 < c - b^2$, and the new vertex is below P .

Therefore the correct answer is F.

Question 15

From the question, it seems that we are expected to work out the first few values of the functions $f(n)$ and $g(n)$, and to look for a pattern.

n	$f(n)$	$g(n)$
1	5	3
2	16	8
3	8	4
4	4	2
5	2	1
6	1	6
7	4	3
8	2	8
9	1	4
10	4	2
11	2	1
12	1	6

So after the first three terms, $f(n)$ repeats every 3 terms: for $k \geq 1$, $f(3k+1) = 4$, $f(3k+2) = 2$, $f(3k+3) = 1$, while $g(n)$ repeats every 6 terms, and we can write similar expressions for $g(6k+r)$.

We could either calculate $f(n) - g(n)$ for all of these terms, and find the repeating pattern there, or find $f(1000)$ and $g(1000)$ separately. The latter approach seems like less effort, so we will go for that.

Note that $1000 = 3 \times 333 + 1 = 6 \times 166 + 4$. So $f(1000) = 4$, $g(1000) = 2$.

Thus $f(1000) - g(1000) = 4 - 2 = 2$ and the answer is D.

Question 16

For a function to be a counterexample to (*), it must be the case that $f(x)$ is an integer for every integer x , but for which $f'(x)$ is **not** an integer for every integer x .

We can start by substituting in $x = 0$ and $x = 1$ into the given functions to see whether $f(x)$ is even potentially an integer for every integer x :

A $f(0) = \frac{1}{4}$. **This cannot be a counterexample.**

B $f(0) = 0$, $f(1) = \frac{3}{2}$. **This cannot be a counterexample.**

C $f(0) = 0$, $f(1) = 2$. **This could potentially be a counterexample.**

D $f(0) = 0$, $f(1) = 1$. **This could potentially be a counterexample.**

So we now only need to consider C and D.

Our next step is to differentiate them and then substitute $x = 0$ and $x = 1$ into the derivatives.

C $f'(x) = \frac{4x^3 + 3x^2 + 2x + 1}{2}$, $f'(0) = \frac{1}{2}$, so this is potentially a counterexample.

D $f'(x) = \frac{4x^3 + 6x^2 + 2x}{4}$, $f'(0) = 0$, $f'(1) = 4$.

It therefore seems most likely that C is a counterexample and D is not. To prove this, we need to show that for C, $f(x)$ is an integer for every integer x , and that for D, either $f(x)$ is not an integer for some integer x or that $f'(x)$ is an integer for every integer x .

For the purposes of the test, it probably makes most sense at this point to choose the answer C and move on, but we will go on here to complete the solution.

For C, we have

$$f(x) = \frac{x(x^3 + x^2 + x + 1)}{2} = \frac{x(x+1)(x^2+1)}{2}$$

We see that $x(x+1)$ is even if x is even and also if x is odd, so $x(x+1)(x^2+1)$ is even for every integer x , hence $f(x)$ is an integer for every integer x . Therefore C is a counterexample.

For D, we have

$$f(x) = \frac{x^2(x^2 + 2x + 1)}{4} = \frac{x^2(x+1)^2}{4}$$

Since $x(x+1)$ is even for every integer x , as we just showed, $x^2(x+1)^2$ is a multiple of 4 for every integer x , and thus $f(x)$ is an integer for every integer x .

For the derivative,

$$f'(x) = \frac{x(2x^2 + 3x + 1)}{2} = \frac{x(x+1)(2x+1)}{2}$$

and as before, the numerator is always even, so $f'(x)$ is an integer for every integer x . Therefore D is not a counterexample.

Question 17

We begin with the original statement with S replaced by T :

T is stapled **if and only if for every** whole number a which is in T , **there exists** a prime factor of a which divides **at least one** other number in T .

Therefore, negating this:

T is **not** stapled **if and only if** it is **not** the case that **for every** whole number a which is in T , **there exists** a prime factor of a which divides **at least one** other number in T .

We can negate the outer ‘**for every**’ to get ‘**there exists** ... such that it is **not** the case that ...’:

T is **not** stapled **if and only if there exists** a whole number a which is in T such that it is **not** the case that **there exists** a prime factor of a which divides **at least one** other number in T .

So the answer is one of E, F, G, H.

Our current phrasing is ‘it is **not** the case that **there exists**’; we could convert this into ‘**for every** ...’, it is **not** the case that ...’, but none of these options use this wording. Instead, E, F and G use the wording ‘there is no prime factor’, which is the same as ‘there does not exist a prime factor’, which in turn is the same as ‘it is not the case that there exists a prime factor’. So we can rewrite the previous statement as:

T is **not** stapled **if and only if there exists** a whole number a which is in T such that there is no prime factor of a which divides **at least one** other number in T .

Hence the correct answer is F.

Question 18

We could first check to see whether the answer makes sense. If we substitute $n = 1$ into the original equation, we get $\frac{1}{4} < \left(\frac{1}{32}\right)^{10}$, which is clearly nonsense, so there has to be a mistake somewhere in the argument.

There are two straightforward approaches to finding the mistake: one is to check that the logic of each step works, the other is to try a value of n which either does or does not solve the original inequality, and check that each step works for this value of n . Since we know that the solution is wrong, it must fail somewhere, at least when $n = 1$.

Approach 1: checking the logic

Clearly step (V) is correct, so the mistake must be earlier.

Step (I) applies the same function to both sides of an inequality, so it looks fine.

Step (II) is just a simplification of the logarithm expressions using the power rule. The powers are positive integers, and both $\frac{1}{4}$ and $\frac{1}{32}$ are positive, so this step is fine.

Starting from the other end, step (IV) is just a calculation, and we can check quite easily that it is correct.

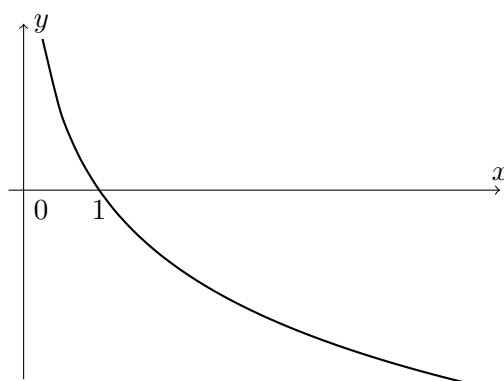
Step (III) looks like it is probably wrong, as we are performing a division with an inequality.

However, with more care, we can calculate $\log_{\frac{1}{2}}\left(\frac{1}{4}\right) = 2$ (as we need to do to check step (IV) anyway), so step (III) is just dividing by 2, and is actually fine.

Unfortunately, though, every step looks fine.

Working through them once more, steps (IV) and (V) are certainly fine, as is step (III). We have argued that step (II) is correct, so by elimination it must be step (I) which is wrong.

It is helpful to sketch a graph of $y = \log_{\frac{1}{2}} x$ to understand why step (I) is wrong. We have $\log_{\frac{1}{2}}\left(\frac{1}{2}\right) = 1$, $\log_{\frac{1}{2}} 1 = 0$, $\log_{\frac{1}{2}} 2 = -1$, so the graph looks like this:



So $\log_{\frac{1}{2}} x$ is a *decreasing* function: if $x < y$, then $\log_{\frac{1}{2}} x > \log_{\frac{1}{2}} y$; in other words, $\log_{\frac{1}{2}}$ reverses the direction of an inequality. Therefore step (I) is incorrect: the inequality should have been reversed in this step.

The correct answer is therefore B.

Approach 2: substituting a value of n

We have already shown that $n = 1$ does not satisfy the original inequality, yet it is given as a solution at the end of step (V). So we use this value of n to find at least one error in the argument.

Note that this approach is not guaranteed to work: there may be other errors in the argument that will escape detection by this method, but given that the options tell us that only one step is invalid, it will be sufficient for our purposes.

It will help us with our calculations if we write everything in terms of powers of $\frac{1}{2}$.

The original inequality can be written as

$$\left(\frac{1}{2}\right)^{2n} < \left(\frac{1}{2}\right)^{50}$$

Substituting $n = 1$ gives

$$\left(\frac{1}{2}\right)^2 < \left(\frac{1}{2}\right)^{50}$$

which is *not* true.

After step (I), the inequality reads

$$\log_{\frac{1}{2}} \left(\frac{1}{2}\right)^{2n} < \log_{\frac{1}{2}} \left(\frac{1}{2}\right)^{50}$$

Substituting $n = 1$ gives

$$\log_{\frac{1}{2}} \left(\frac{1}{2}\right)^2 < \log_{\frac{1}{2}} \left(\frac{1}{2}\right)^{50}$$

which we can simplify to

$$2 < 50$$

which *is* true.

So something has gone wrong in step (I), and this is where the error lies.

Commentary: There is a slightly subtle question about whether the student intended each of the steps in this argument to be reversible. For example, if a student started with the statement $x + 5 = 8$ and deduced $x > 0$, they have made a correct deduction (though not ‘solved the equation’), but the argument is not reversible: $x > 0$ does not imply that $x + 5 = 8$. So we will assume that in this case they intend to say that $1 \leq n \leq 24$ are all solutions to the original equation.

It follows that step (I) is where the mistake lies, as $n = 1$ satisfies the inequality after step (I) but not the inequality before it.

If we did not make this assumption about the student’s intentions, we could repeat the argument above with a value of n which does satisfy the original inequality, such as $n = 100$; then the first inequality is true but the second inequality, obtained after step (I), is false, so the proposed implication is false.

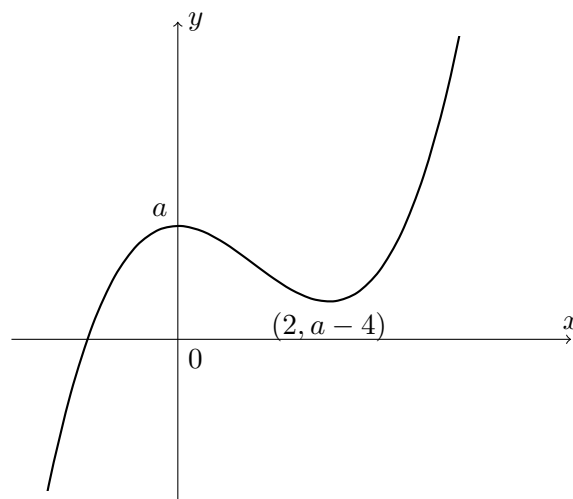
Question 19

We recall that a '**sufficient** condition' means that if the condition holds, then the equation has exactly one real root.

We can start by sketching the graph of the cubic $y = x^3 - 3x^2 + a$, noting that the impact of changing a is just to translate the graph vertically.

We differentiate to find the stationary points: $\frac{dy}{dx} = 3x^2 - 6x$, so $\frac{dy}{dx} = 0$ when $x = 0$ and $x = 2$, so the stationary points are at $(0, a)$ and $(2, a - 4)$.

So the graph looks like this:



There will be one real root for the equation $x^3 - 3x^2 + a = 0$ if either the local maximum lies below the x -axis or if the local minimum lies above the x -axis, as in the case sketched.

The first occurs when $a < 0$, the second when $a - 4 > 0$ or equivalently, when $a > 4$.

So a **necessary and sufficient** condition is $a < 0$ or $a > 4$. This does not, however, appear as one of the options, so we need to go through them to see which one is **sufficient**. (Recall from earlier that **sufficient** means that if this condition holds, then the equation has exactly one real root, but not necessarily the other way round.)

- A** This includes $a = 1$, which gives more than one real root.
- B** This includes $a = 0$, which gives more than one real root (as the minimum lies on the x -axis).
- C** This includes $a = 4$, which gives more than one real root (as the maximum lies on the x -axis).
- D** This includes $a = 1$ again.
- E** This is $a > 4$ or $a < -4$, both of which are part of the valid range, so this is a **sufficient** condition.
- F** This includes $a = 1$ again.
- G** This gives more than one real root.
- H** $a = \frac{3}{2}$ gives more than one real root.

So the correct answer is E.

Question 20

We can write this out in a grid to see what the current state of knowledge is.

	Position				
	1	2	3	4	5
Tried	a	b	c	d	e
	c	d	b	e	a
	e	a	d	b	c
Possible	b	c	a	a	b
	d	e	e	c	d

So positions 1 and 5 contain **b** and **d** in some order, while positions 2, 3, 4 contain either **cea** or **eac** in that order.

We certainly do not know the correct password yet.

We could sensibly try **bcead** as our next attempt. What might we be told in response?

If we have the **b** and **d** in the correct order, these 2 letters will be correct, otherwise neither is correct.

If we have the middle three letters **cea** in the correct order, then these 3 letters will be correct, otherwise none of them is.

So:

- if both are correct, the computer will tell us that 5 letters are correct, and we are done;
- if the outer pair are correct but the middle ones are not, the computer will tell us that 2 letters are correct, and we can deduce that the correct password is **beacd**;
- if the outer pair are wrong but the middle ones are correct, the computer will tell us that 3 letters are correct, and we can deduce that the correct password is **dceab**;
- if both are wrong, the computer will tell us that 0 letters are correct, and we can deduce that the correct password is **deacb**.

Since the four cases give four distinct responses by the computer (0, 2, 3, 5 correct letters), we can distinguish between them, and therefore deduce the correct password with just this one further attempt.

The correct answer is B.

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